Smooth Models

The models defined before were based purely on configuration.

- Assumes instintaneous velocity

Configuration: Variables needed to define all points on vehicle

State! Append derivatives (accelerations)

↳ control higher order terms for smoother responses.

I Smooth Unicycle Model

$x_1 = x \quad x_2 = y \quad x_3 = \theta \quad x_4 = v \quad x_5 = \omega$

$$\Rightarrow \quad \dot{x}_1 = \dot{x} = v\cos\theta = x_4 \cos x_3$$
$$\dot{x}_2 = \dot{y} = x_4 \sin x_3$$
$$\dot{x}_3 = \dot{\theta} = \omega = x_5$$
$$\left.\begin{aligned} \dot{x}_4 &= \dot{v} = a \\ \dot{x}_5 &= \alpha \end{aligned}\right\} \text{These are inputs} \qquad \begin{aligned} u_1 &= a \\ u_2 &= \alpha \end{aligned}$$

$$\Rightarrow \quad \dot{x} = \begin{bmatrix} x_4 \cos x_3 \\ x_4 \sin x_3 \\ x_5 \\ u_1 \\ u_2 \end{bmatrix}$$

II Smooth Differential Drive

$\dot{\mu}_r = \alpha_r \quad \dot{\mu}_l = \alpha_l$ ← inputs

$x_1 = x \quad x_2 = y \quad x_3 = \theta \quad x_4 = \mu_r \quad x_5 = \mu_l$

$\dot{x}_1 = \frac{r}{2}(x_4 + x_5)\cos x_3 \qquad x_2 = \frac{r}{2}(x_4 + x_5)\sin x_3$

$\dot{x}_3 = r/L\,(x_4 - x_5) \qquad \dot{x}_4 = u_1 = \alpha_r \quad \dot{x}_5 = u_2 = \alpha_l$

II Continuous Steering

$x_1 = x \quad x_2 = y \quad x_3 = \theta \quad x_4 = v \quad x_5 = \phi$

$\dot{x}_1 = x_4 \cos x_3 \qquad \dot{x}_2 = x_4 \sin x_3$

$\dot{x}_3 = \frac{x_4}{L} \tan(x_5) \qquad \dot{x}_4 = u_1 \quad \dot{x}_5 = u_2$

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